

Class Imbalance in Histopathology: Expanding BRACS’s Class Separation Does Not Reduce Its Imbalance Damage

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Abstract

At an imbalance ratio of $\rho = 100$, class imbalance costs BRACS 7.65 points of patient-macro balanced accuracy and TCGA-UT 2.91 points. This report tests whether BRACS’s poorer class separation causes part of that 4.74-point gap. A label-dependent feature intervention moves class centres apart on BRACS and together on TCGA-UT, and four refitted training arms separate prior-only, support-only, and total damage. A pre-registered pilot on one training draw per split passed the reproduction, manipulation, precision, and numerics gates, but failed the mechanism gate on both datasets. Expanding BRACS’s separation raised balanced-arm validation accuracy by 24.7 points, yet total imbalance damage rose by 2.95 points instead of falling. Contracting TCGA-UT’s separation raised total damage by only 1.51 points, below the 2-point threshold. The main experiment was therefore not submitted, and no causal claim about separation is made. One exploratory pattern is consistent across both datasets: prior-only damage moved in the predicted direction (BRACS -3.72 , TCGA-UT $+1.81$ points), while support-only damage did not.

1 Question and design

The damage studies in this series fit both datasets with the same allocator, splits, class-rank permutations, and logistic readout on frozen Virchow2 features (Queisler, 2026e,d; Zimmermann et al., 2024). BRACS still loses far more balanced accuracy under imbalance. A companion report tried to explain the prior-channel part of this gap by injecting the realized class prior into stored balanced-model decisions; that proxy overshot fitted damage three- to ninefold, and its gate failed (Queisler, 2026c). BRACS’s seven lesion subtypes are related categories on one tissue axis (Brancati et al., 2022), so its classes may sit closer together in feature space than TCGA-UT’s thirty tumour types. If so, the same prior shift and the same thin tail support would flip more decisions on BRACS.

The intervention changes only the class locations:

$$x^{(\alpha)} = x + (\alpha - 1) (\mu_y - \bar{\mu}), \quad (1)$$

where μ_y is the training-only, patient-balanced centre of class y and $\bar{\mu}$ is the unweighted mean of the class centres. Labels, patients, patch allocations, class ranks, and within-class variation stay fixed. The same shift applies to training, validation, and test features. The intervention uses the true label at evaluation time, so it tests a mechanism inside the representation, not a deployable method.

The strength α was fixed before any refit. The *separation index* J is the median nearest-class-centre distance divided by the pooled within-class RMS deviation. BRACS’s expansion is $\alpha = J_{\text{TCGA-UT}}/J_{\text{BRACS}}$ per split, and TCGA-UT’s contraction is its reciprocal. J was 0.44–0.48 on BRACS and 0.68–0.71 on TCGA-UT, which gives $\alpha = 1.43$ – 1.55 . This lies inside the pre-registered range $[1.25, 4]$ in every split, so the precheck passed.

Both datasets use 10 patients per class and 32 balanced patches per patient. Each setting has four arms at $\rho = 100$: balanced rows and prior (B), balanced rows with an imbalanced prior (P), imbalanced rows reweighted to a balanced prior (S), and imbalanced rows and prior (R) (Queisler, 2026b,a). Damage is $D_P = A_B - A_P$, $D_S = A_B - A_S$, and $D_R = A_B - A_R$ on patient-macro balanced accuracy. The pilot fit expanded BRACS, native TCGA-UT, and contracted TCGA-UT, plus a native BRACS replay, on draw 0 of all three frozen splits (48 fits).

2 Pilot outcome

Table 1 lists the five gates. The pre-registered rule stops the main experiment if any gate fails, so the 144 main-experiment fits were not submitted and no alternative strengths were searched.

Gate	BRACS (expanded)	TCGA-UT (contracted)	Result
1 Reproduction	native replay matches stored test BA, max. difference 0.00 pp	not applicable	pass
2 Manipulation	B-arm validation BA +24.7 pp (splits +27.0, +24.0, +23.2)	B-arm validation BA −15.8 pp (splits −16.4, −14.8, −16.2)	pass
3 Mechanism	$\Delta D_R = +2.95$ pp; required ≤ -2	$\Delta D_R = +1.51$ pp; required $\geq +2$	fail
4 Precision	projected power 0.95	projected power 1.00	pass
5 Numerics	all converged; 5 fits at lower λ bound	all converged	pass

Table 1: Pilot gates on validation data, draw 0, three splits. Δ is the treated setting minus its native control at 10 patients per class. Gate 3 fails on both datasets, so the main experiment stops.

The manipulation worked as intended. Expansion made BRACS much easier and contraction made TCGA-UT harder, and all balanced accuracies stayed inside the non-trivial range of 25–90 percent. The failure is therefore not a weak intervention. The mechanism gate failed because damage did not follow the change in separation (Table 2). On BRACS, total damage rose in every split (+5.70, +2.81, +0.35 points), which is the opposite of the predicted rescue. On TCGA-UT, total damage rose in two of three splits (+3.43, −0.09, +1.21 points), but the pooled change of 1.51 points missed the 2-point threshold.

Setting	ΔD_P	ΔD_S	ΔD_R	Predicted sign
BRACS, expanded vs. native	−3.72	+6.50	+2.95	−
TCGA-UT, contracted vs. native	+1.81	−0.04	+1.51	+

Table 2: Pooled change in validation damage (percentage points) between treated and native settings. Only the prior channel D_P moves in the predicted direction on both datasets; the support channel D_S does not.

Two gates that passed need qualification. The precision gate simulates power at an assumed 2-point effect from the spread of only three split residuals, and it does not correct for the two primary comparisons as planned; its pass matters only if gate 3 had passed. The numerics gate flags five BRACS fits, three of them in the expanded setting, whose selected regularization sits at the weakest grid value $\lambda = 10^{-6}$. Expanded features may prefer weaker regularization than the grid allows.

3 Interpretation

Under this operational definition of separation, the pilot does not support separation as a cause of BRACS’s larger imbalance damage. Moving BRACS’s class centres to TCGA-UT’s relative

spacing did not reduce total damage, even though it raised balanced accuracy by about 25 points.

The channel split in Table 2 suggests why, but it comes from one draw on validation data with no uncertainty interval, so it is exploratory. Prior-only damage behaved as the separation hypothesis predicts on both datasets: it fell when BRACS classes moved apart and rose when TCGA-UT classes moved together. Support-only damage did not. On BRACS it rose by 6.50 points, more than enough to cancel the prior-channel gain. One candidate explanation is headroom: a better-separated balanced model classifies more tail patients correctly, so it has more to lose when tail support thins. The BRACS tail-assignment study found that headroom dominates tail damage (Queisler, 2026f). This pilot did not test that explanation.

Three limits apply. The intervention is an oracle that uses true labels, so it says nothing about tissue biology or a deployable remedy. J is one summary of class geometry; another definition could give a different α . The two datasets still differ in class count (seven against thirty), which support matching does not control. A follow-up that targets the prior channel alone would need its own pre-registered gate; the present gate cannot be reused after seeing these outcomes.

References

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